Predictive Football Analytics Using Machine Learning

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Does destination-team context add predictive value beyond player characteristics?

Academic project

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Research motivation

- Player performance after a transfer is uncertain.
- The destination club may shape opportunities, tactical role, and the environment in which a player performs.
- Player-only models may therefore omit useful contextual information.

Central research question

Does adding destination-team context improve prediction of a player's post-transfer goals and assists per 90 minutes?



Conceptual research pathway

Aim and objectives

Aim

Evaluate whether destination-team context improves machine-learning predictions of post-transfer attacking performance.

1. Establish a player-characteristics baseline.
2. Add basic destination-team context and compare performance.
3. Test whether richer team and tactical context provides further improvement.
4. Use paired error comparisons and uncertainty intervals to assess whether observed differences are convincing.

Scope

The current review focuses on the existing A/B/C experiments and the separate corrected audit. Proposed future modules are not treated as completed work.

Prediction target and evaluation

Target variable

$$\texttt{post_GA90} = \frac{\text{post-transfer goals} + \text{assists}}{\text{minutes played}} \times 90$$

The target represents attacking output per 90 minutes after the transfer.

Evaluation metrics

- **MAE**: average absolute prediction error; lower is better.
- **RMSE**: penalises larger errors more strongly; lower is better.
- R^2: proportion of target variance explained relative to a mean baseline; higher is better.

All reported results are regression metrics. An R^2 value is **not** classification accuracy.

Model comparison design

Model	Input strategy	Purpose
Model A	Player characteristics	Baseline: estimate performance from player-level information.
Model B	Player characteristics + basic destination-team context	Test whether basic team context adds predictive value.
Model C	Player characteristics + richer team/tactical context	Test whether expanded context improves on Model B.

Fair comparison

The main A/B/C prediction files contain **1,194 aligned test records per model**. The paired comparisons use corresponding test cases rather than unrelated samples.

Main results: model metrics

Model	Test N	MAE	RMSE	R^2
A: Player only	1,194	0.123159	0.172344	0.551043
B: + Basic team context	1,194	0.121047	0.169027	0.568156
C: + Richer context	1,194	0.121325	0.169764	0.564382

Model B vs A

MAE falls by approximately **1.71%**; RMSE also decreases and R^2 increases.

Model C vs B

Model C does not improve on Model B on these main test results.

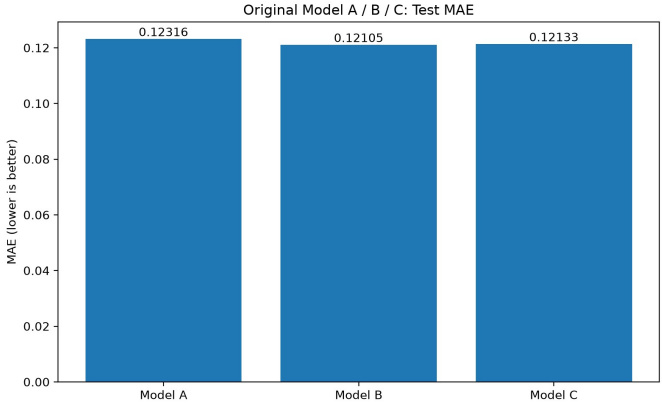
Main results: terminal comparison

```
[1] ORIGINAL MODEL A / B / C
  Model  Test_N      MAE     RMSE       R2  MAE_improvement_vs_A_pct
Model A    1194 0.123159 0.172344 0.551043                  0.000000
Model B    1194 0.121047 0.169027 0.568156                  1.714856
Model C    1194 0.121325 0.169764 0.564382                  1.489132

[2] CORRECTED AUDIT (SEPARATE TEST SET)
                  Variant  Test_N      MAE     RMSE       R2  MAE_improvement_vs_core_pct
                     Core    1223 0.120966 0.170774 0.554984                     0.000000
                   Mode A    1223 0.119102 0.167612 0.571309                     1.540929
            Full enhanced    1223 0.118346 0.166695 0.575988                     2.165898
 Without tactical/manager    1223 0.118345 0.166290 0.578044                     2.166725
Without squad competition    1223 0.118494 0.166298 0.578005                     2.043549
            Without coach    1223 0.119102 0.167612 0.571309                     1.540929
```

Figure 1. Terminal output showing the three-model metric comparison.

Main results: test MAE across A, B, and C



Interpretation

Model B has the lowest MAE in the main comparison. Model C's MAE is slightly higher than Model B's, so richer context does not automatically translate into better predictions.

Statistical evidence: paired test errors

Comparison	MAE difference*	Wilcoxon p	Interpretation
A vs B	0.002112	0.001579	Evidence that B reduces error relative to A.
A vs C	0.001834	0.019487	Evidence that C reduces error relative to A.
B vs C	-0.000278	0.422961	No convincing evidence of a difference between B and C.

Bootstrap 95% confidence intervals for paired MAE difference

- A–B: [0.000764, 0.003406] — excludes zero.
- A–C: [0.000192, 0.003431] — excludes zero.
- B–C: [−0.001564, 0.001006] — includes zero.

* Difference is defined as MAE of the first model minus MAE of the second; positive values favour the second model.

Which inputs matter? Random Forest feature importance

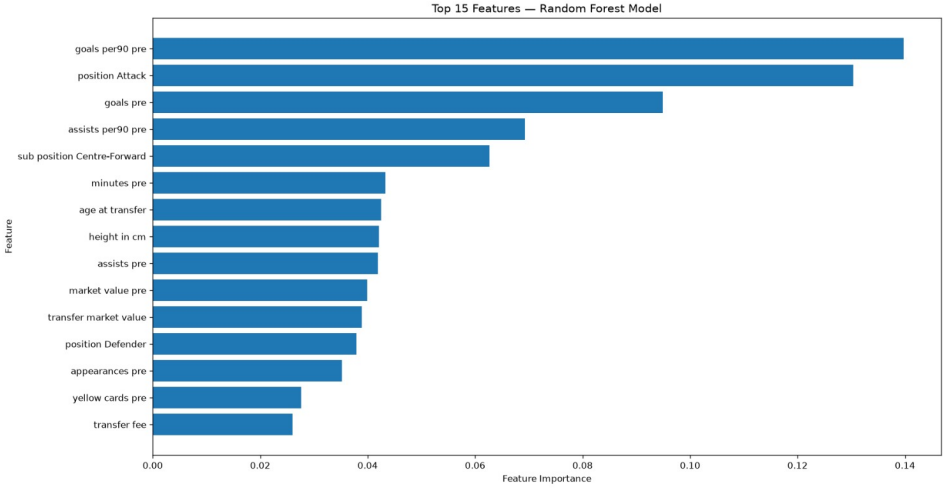


Figure 3. Top 15 features from the Random Forest model. Feature importance indicates model reliance, not causality.

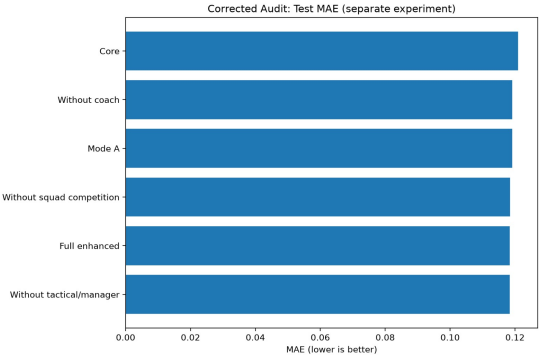
Separate experiment: corrected audit

Keep this result separate from the main A/B/C comparison

The corrected audit uses a different evaluation cohort and reports $N = 1{,}223$. Its values should not be merged with the 1,194-record main comparison.

Audit configuration	N	MAE	RMSE	R^2
Core	1,223	0.120966	0.170774	0.554984
Mode A	1,223	0.119102	0.167612	0.571309
Full enhanced	1,223	0.118346	0.166695	0.575988
Without tactical/manager	1,223	0.118345	0.166290	0.578044
Without squad competition	1,223	0.118494	0.166298	0.578005
Without coach	1,223	0.119102	0.167612	0.571309

Separate experiment: corrected audit MAE



Interpretation caution

This is a separate audit experiment, not the same test set as the main A/B/C analysis. It is useful as a secondary check, but direct numerical comparisons across the two experiments are not warranted without a shared protocol and cohort.

Limitations and interpretation

- The improvement from Model A to Model B is statistically supported in the supplied paired comparisons, but its absolute size is modest.
- Model C does not significantly outperform Model B; additional features may add complexity without reliable gains.
- Feature importance is descriptive and should not be interpreted as evidence that a feature causes performance.
- Predictions of post-transfer performance are affected by factors that may not be captured in the available data, including playing time, role changes, injuries, and adaptation.
- Results depend on the current dataset, target definition, feature construction, and evaluation split.

Conclusion and next steps

Main conclusion

In the main 1,194-record comparison, adding basic destination-team context improves predictive performance over the player-only baseline. The richer Model C does not show a statistically convincing improvement over Model B.

Next steps

1. Confirm that all feature construction and model comparisons are reproducible from the final scripts.
2. Document the train/test split and any preprocessing choices.
3. Validate results on a future-time or external cohort if feasible.
4. Verify all literature metadata against the original publications before final submission.

Thank you!!!!